IBM CONSULTING

2026 ICA BOB-A-THON · USE CASE PROPOSAL

Spec-Driven AI Development with BMad + Bob

A structured four-agent workflow that turns Bob from a code generator into a disciplined engineering partner.

Proposed by	Category	Tools	Year
ahmed elpannann	Developer Productivity	Bob · MCP · BMad-Method	2026

THE IDEA

Making Bob Think Before It Codes

Most AI coding assistants — including Bob — are reactive by default: you describe a feature and they immediately generate code. This produces fast first drafts but creates a hidden cost. Without documented requirements, a technical design, or a task breakdown, the AI has no shared understanding of what it is building, why specific decisions were made, or how the work connects to the broader system. The result is code that is often correct in isolation but misaligned with the product, difficult to review, and expensive to change.

This proposal introduces a **Spec-Driven MCP server** layered with the **BMad-Method** agent framework, running entirely inside Bob. The MCP server exposes fourteen structured tools that enforce a three-phase workflow — Requirements, Design, Tasks — before any implementation begins. On top of this, four specialist agent personas (Mary the Analyst, Winston the Architect, John the Product Manager, and Amelia the Developer) guide the user through each phase in natural language, calling the MCP tools to persist every decision as versioned Markdown files in the project repository. The result is a complete, auditable engineering record — requirements written in EARS notation, a design document with Architecture Decision Records, and a task list with acceptance criteria — all generated collaboratively with Bob and stored alongside the code. Implementation only begins once all three documents are locked, and task status is tracked in real time via MCP. The server also includes a privacy filter that automatically redacts API keys and secrets from any spec file before it is written to disk.

The workflow is compatible with Bob, Cline, Cursor, and Windsurf with no changes to the server. It requires no new infrastructure — only a Python virtual environment and a single JSON configuration file pointing Bob at the MCP server.

Traceable Decisions
Every ADR and requirement is versioned in the repo

Zero Rework
Design is locked before a line of code is written

Privacy by Default
Secrets auto-redacted from all spec files

Any AI Assistant
Works with Bob, Cline, Cursor, Windsurf

WORKFLOW ARCHITECTURE — HOW IT WORKS

You (Developer)

Bob — BMad Agent

MCP Server Tool Call

Persisted Spec File (.specs/)

Agent Handoff